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Q1. Directory Creation

Create the following directory structure in your home folder:

Parallel_Programming/

├── OpenMP/

├── MPI/

├── CUDA/

└── Scripts/

```
Welcome to Cloud Shell! Type "help" to get started.
To set your Cloud Platform project in this session use `gcloud config set project [PROJE
CT ID]`.
You can view your projects by running `gcloud projects list`.
armaanpatnaik2004@cloudshell:~$ echo "Question 1 - Directory Creation"
Question 1 - Directory Creation
armaanpatnaik2004@cloudshell:~$ mkdir -p ~/Parallel Programming/{OpenMP,MPI,CUDA,Scripts
}
armaanpatnaik2004@cloudshell:~$ ls -R ~/Parallel Programming
/home/armaanpatnaik2004/Parallel Programming:
CUDA  MPI  OpenMP  Scripts

/home/armaanpatnaik2004/Parallel Programming/CUDA:

/home/armaanpatnaik2004/Parallel Programming/MPI:

/home/armaanpatnaik2004/Parallel Programming/OpenMP:

/home/armaanpatnaik2004/Parallel Programming/Scripts:
armaanpatnaik2004@cloudshell:~$
```

Q.2 File Creation

Inside each directory, create the following files:

OpenMP/

omp_hello.c

MPI/

mpi_hello.c

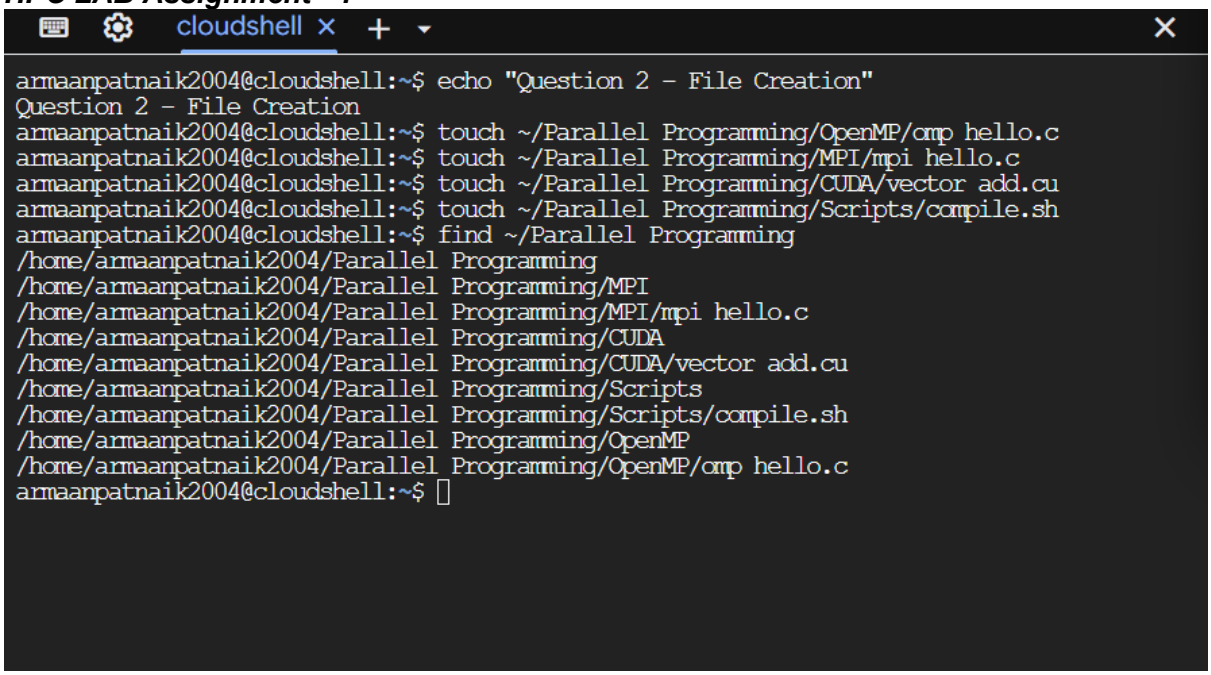
CUDA/

vector_add.cu

Scripts/

compile.sh

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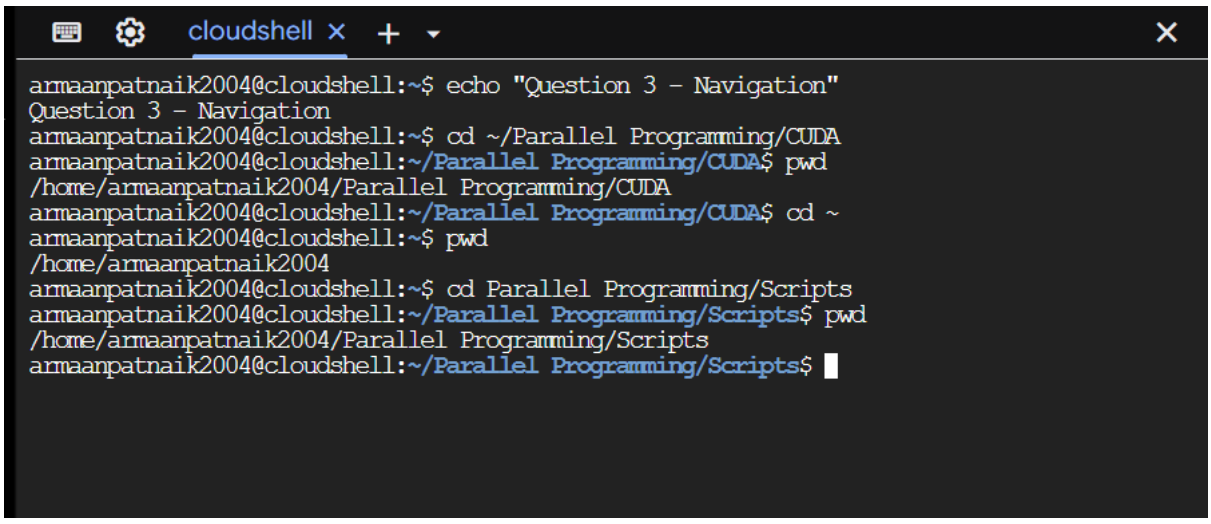
Q.3 - Navigation

Without using the mouse,

move into the CUDA directory

return to the home directory

move to the Scripts directory using a relative path



Q4. Listing Files

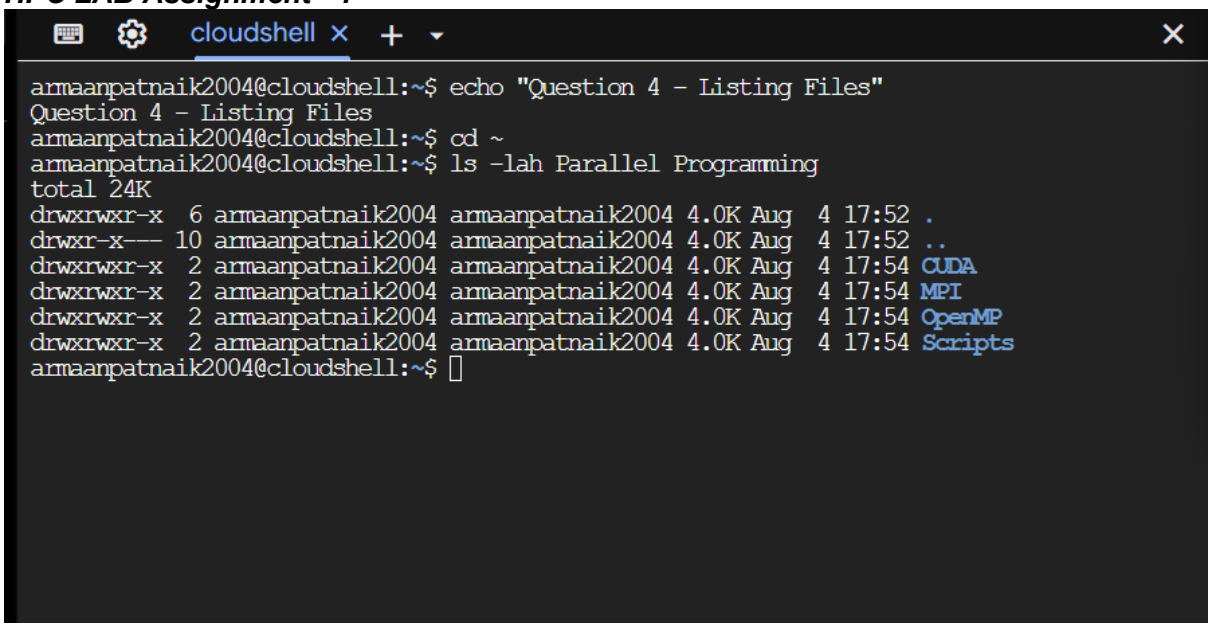
Display

all files

hidden files

file sizes in human-readable format inside the Parallel_Programming directory.

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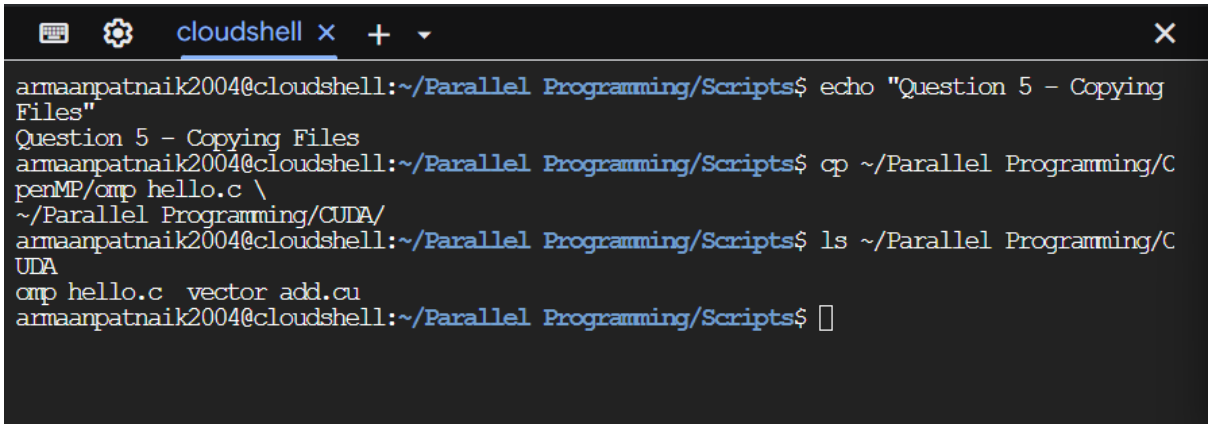


Q.5- Copying Files

Copy

omp_hello.c

from the OpenMP folder into the CUDA folder.



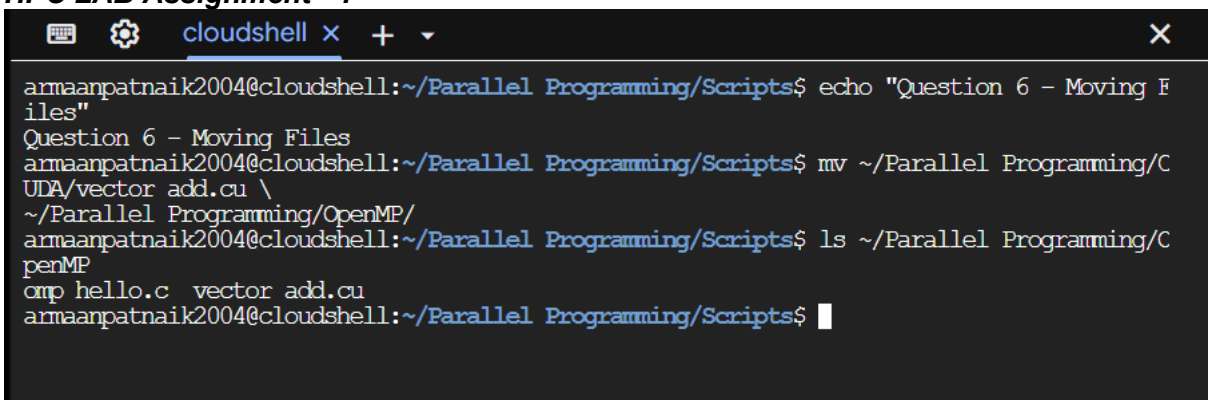
Q.6. Moving Files

Move

vector_add.cu

from CUDA into OpenMP.

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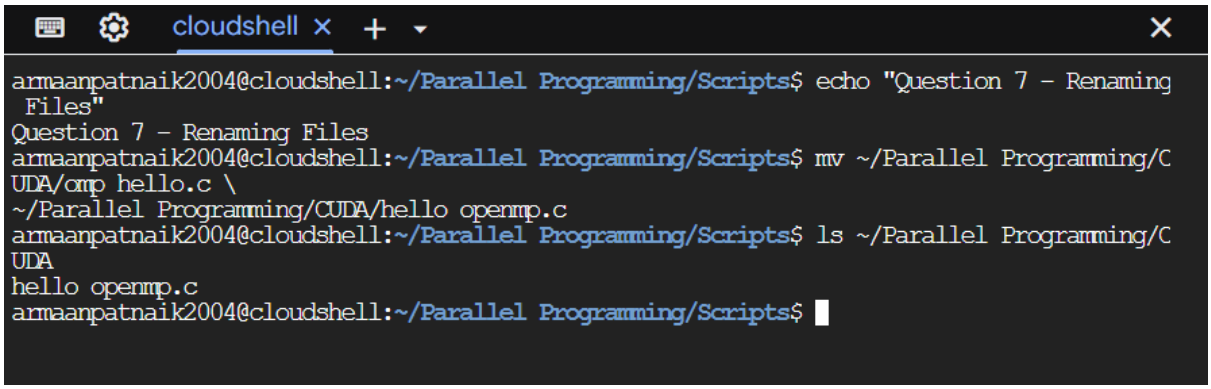


Q.7. Renaming Files

Rename

omp_hello.c

to hello_openmp.c

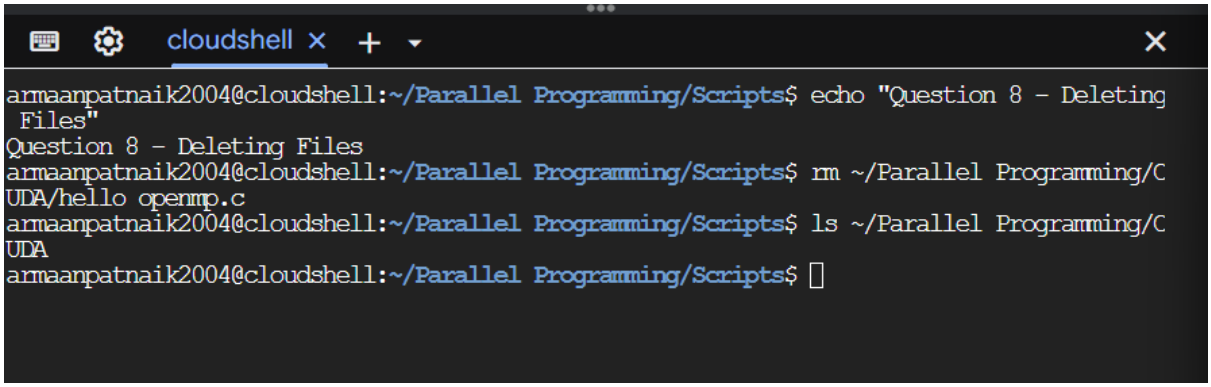


Q8. Deleting Files

Delete

hello_openmp.c

without deleting the directory.



Q9. Searching

Find every file having the extension .c inside the Parallel_Programming directory.

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```
cloudshell ×
armaanpatnaik2004@cloudshell:~/Parallel Programming/Scripts$ echo "Question 9 - Searchin
g"
Question 9 - Searching
armaanpatnaik2004@cloudshell:~/Parallel Programming/Scripts$ find ~/Parallel Programming
 -name "*.c"
/home/armaanpatnaik2004/Parallel Programming/MPI/mpi hello.c
/home/armaanpatnaik2004/Parallel Programming/OpenMP/omp hello.c
armaanpatnaik2004@cloudshell:~/Parallel Programming/Scripts$
```

Q.10. Finding Current Location

Display the current working directory.

```
cloudshell ×
armaanpatnaik2004@cloudshell:~/Parallel Programming/Scripts$ echo "Question 10 - Current
 Working Directory"
Question 10 - Current Working Directory
armaanpatnaik2004@cloudshell:~/Parallel Programming/Scripts$ pwd
/home/armaanpatnaik2004/Parallel Programming/Scripts
armaanpatnaik2004@cloudshell:~/Parallel Programming/Scripts$
```

Q.11. Viewing File Contents

Create a text file containing five Linux commands and display its contents using

Cat more less head tail

```
cloudshell ×
armaanpatnaik2004@cloudshell:~/Parallel Programming/Scripts$ echo "Question 11 - Viewing
 File Contents"
Question 11 - Viewing File Contents
armaanpatnaik2004@cloudshell:~/Parallel Programming/Scripts$ cat > commands.txt
pwd
ls
mkdir
cp
mvarmaanpatnaik2004@cloudshell:~/Parallel Programming/Scripts$ cat commands.txt
pwd
ls
mkdir
cp
mvarmaanpatnaik2004@cloudshell:~/Parallel Programming/Scripts$ more commands.txt
pwd
ls
mkdir
cp
mv
armaanpatnaik2004@cloudshell:~/Parallel Programming/Scripts$ less commands.txt

[1]+  Stopped                 less commands.txt
armaanpatnaik2004@cloudshell:~/Parallel Programming/Scripts$ less commands.txt
armaanpatnaik2004@cloudshell:~/Parallel Programming/Scripts$ head commands.txt
pwd
ls
mkdir
cp
mvarmaanpatnaik2004@cloudshell:~/Parallel Programming/Scripts$ tail commands.txt
pwd
ls
mkdir
cp
mvarmaanpatnaik2004@cloudshell:~/Parallel Programming/Scripts$ fg
less commands.txt
armaanpatnaik2004@cloudshell:~/Parallel Programming/Scripts$
```

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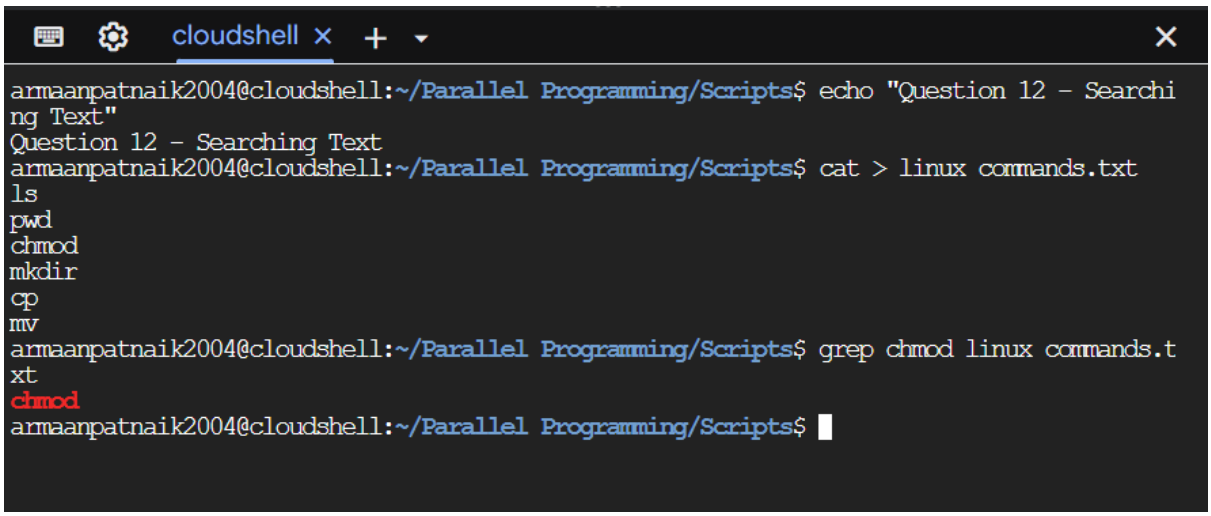
Q.12. Searching Text

Create a file containing names of Linux commands.

Search for

chmod

using an appropriate Linux command.



Q.13. Counting

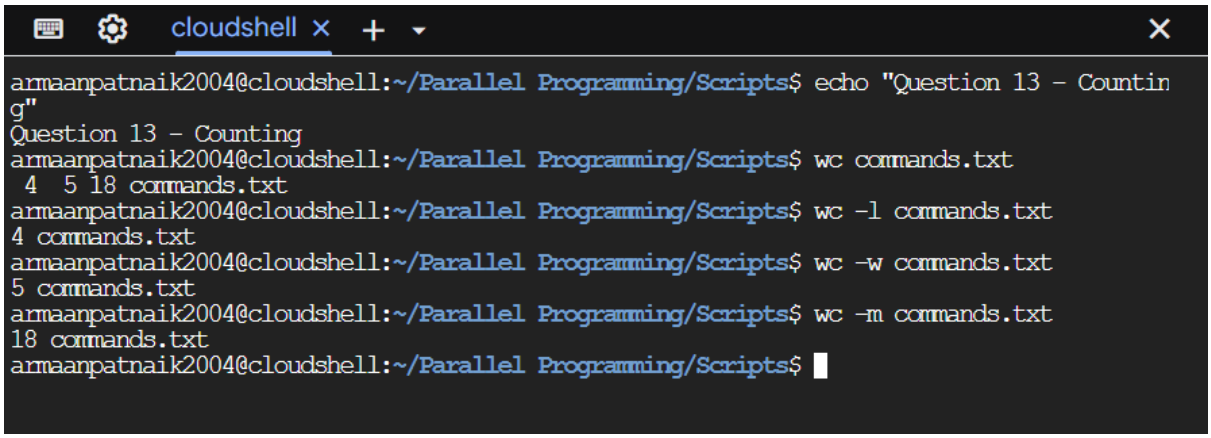
Count

number of lines

number of words

number of characters

in a text file.



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Q.14. File Permissions

Make

compile.sh

executable.

Verify that the permissions have changed.

```
cloudshell ×
armaanpatnaik2004@cloudshell:~/Parallel Programming/Scripts$ echo "Question 14 - File Pe
rmissions"
Question 14 - File Permissions
armaanpatnaik2004@cloudshell:~/Parallel Programming/Scripts$ chmod +x ~/Parallel Program
ming/Scripts/compile.sh
armaanpatnaik2004@cloudshell:~/Parallel Programming/Scripts$ ls -l ~/Parallel Programmin
g/Scripts
total 8
-rw-rw-r-- 1 armaanpatnaik2004 armaanpatnaik2004 18 Aug  4 18:12 commands.txt
-rwxrwxr-x 1 armaanpatnaik2004 armaanpatnaik2004  0 Aug  4 17:54 compile.sh
-rw-rw-r-- 1 armaanpatnaik2004 armaanpatnaik2004 25 Aug  4 18:20 linux commands.txt
armaanpatnaik2004@cloudshell:~/Parallel Programming/Scripts$
```

Q.15. Compression Compress the entire Parallel Programming directory into parallel.tar.gz Extract it into another directory. Shell Scripting Practice

```
cloudshell ×
armaanpatnaik2004@cloudshell:~$ echo "Question 15 - Compression"
Question 15 - Compression
armaanpatnaik2004@cloudshell:~$ tar -czvf parallel.tar.gz Parallel Programming
Parallel Programming/
Parallel Programming/MPI/
Parallel Programming/MPI/mpi hello.c
Parallel Programming/CUDA/
Parallel Programming/Scripts/
Parallel Programming/Scripts/parallel.tar.gz
Parallel Programming/Scripts/compile.sh
Parallel Programming/Scripts/Extracted/
Parallel Programming/Scripts/linux commands.txt
Parallel Programming/Scripts/commands.txt
Parallel Programming/OpenMP/
Parallel Programming/OpenMP/vector add.cu
Parallel Programming/OpenMP/omp hello.c
armaanpatnaik2004@cloudshell:~$ mkdir Extracted
armaanpatnaik2004@cloudshell:~$ tar -xzvf parallel.tar.gz -C Extracted
Parallel Programming/
Parallel Programming/MPI/
Parallel Programming/MPI/mpi hello.c
Parallel Programming/CUDA/
Parallel Programming/Scripts/
Parallel Programming/Scripts/parallel.tar.gz
Parallel Programming/Scripts/compile.sh
Parallel Programming/Scripts/Extracted/
Parallel Programming/Scripts/linux commands.txt
Parallel Programming/Scripts/commands.txt
Parallel Programming/OpenMP/
Parallel Programming/OpenMP/vector add.cu
Parallel Programming/OpenMP/omp hello.c
armaanpatnaik2004@cloudshell:~$ ls Extracted
Parallel Programming
armaanpatnaik2004@cloudshell:~$
```

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Q.16. Hello World Script

Write a shell script that prints

Welcome to Parallel Programming

```
cloudshell ×
armaanpatnaik2004@cloudshell:~$ cd ~
armaanpatnaik2004@cloudshell:~$ pwd
/home/armaanpatnaik2004
armaanpatnaik2004@cloudshell:~$ rm -f hello.sh
armaanpatnaik2004@cloudshell:~$ nano hello.sh
armaanpatnaik2004@cloudshell:~$ cat hello.sh
#!/bin/bash

# Question 16
echo "Welcome to Parallel Programming"
armaanpatnaik2004@cloudshell:~$ chmod +x hello.sh
armaanpatnaik2004@cloudshell:~$ ./hello.sh
Welcome to Parallel Programming
armaanpatnaik2004@cloudshell:~$
```

Q.17. Variables

Write a script that stores your Name Roll Number Branch in variables and prints them.

```
cloudshell ×
armaanpatnaik2004@cloudshell:~$ nano variables.sh
armaanpatnaik2004@cloudshell:~$ nano variables.sh
armaanpatnaik2004@cloudshell:~$ cat variables.sh
#!/bin/bash

# Question 17
# Variables

name="Armaan Patnaik"
roll="2223448"
branch="Computer Science"

echo "Name: $name"
echo "Roll Number: $roll"
echo "Branch: $branch"
armaanpatnaik2004@cloudshell:~$ chmod +x variables.sh
armaanpatnaik2004@cloudshell:~$ chmod +x variables.sh
armaanpatnaik2004@cloudshell:~$ ./variables.sh
Name: Armaan Patnaik
Roll Number: 2223448
Branch: Computer Science
armaanpatnaik2004@cloudshell:~$
```

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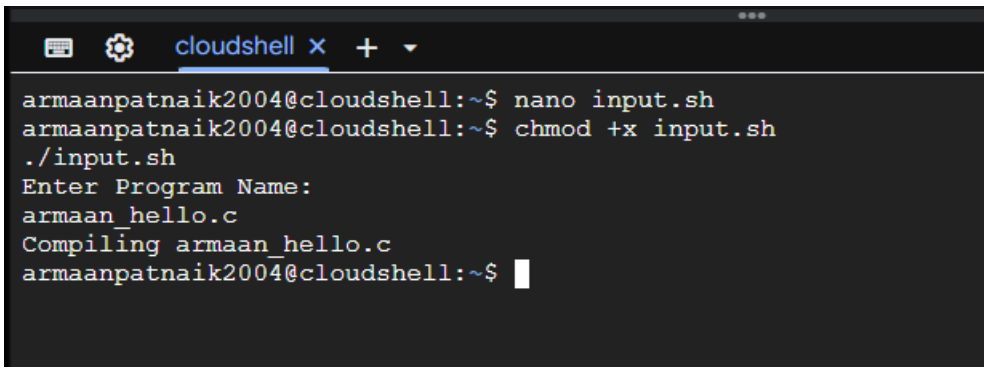
18. User Input

Write a script that asks the user to enter

Program Name

and prints

Compiling <Program Name>



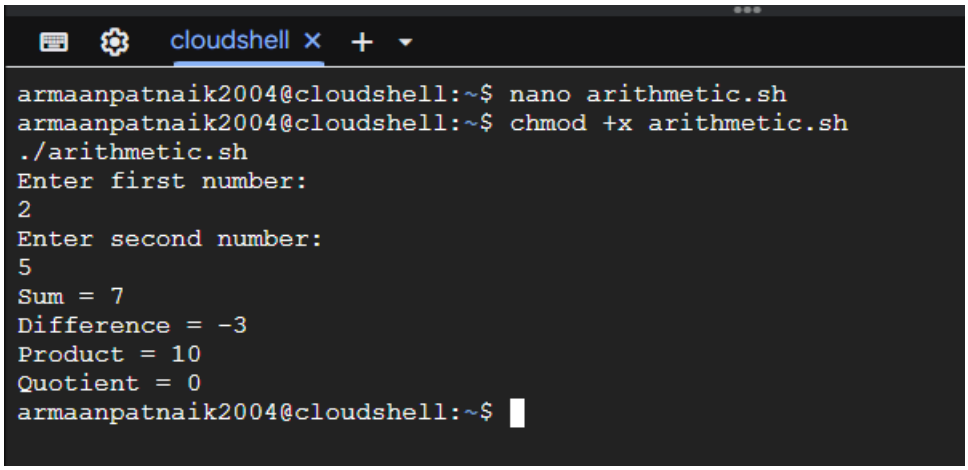
Q.19. Arithmetic

Write a script to input two numbers and display

Sum

Difference

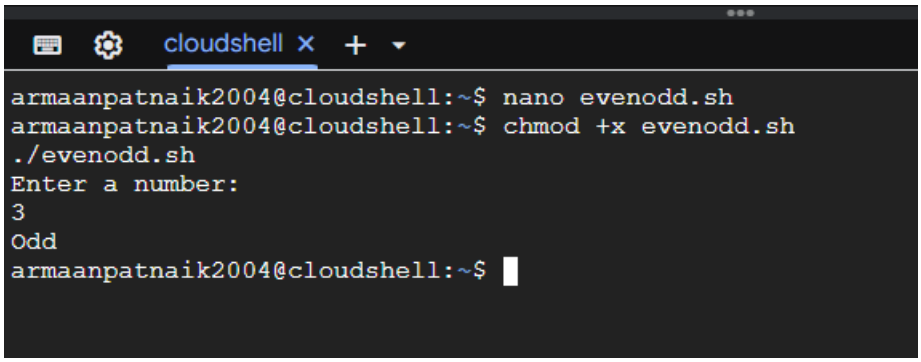
Product Quotient



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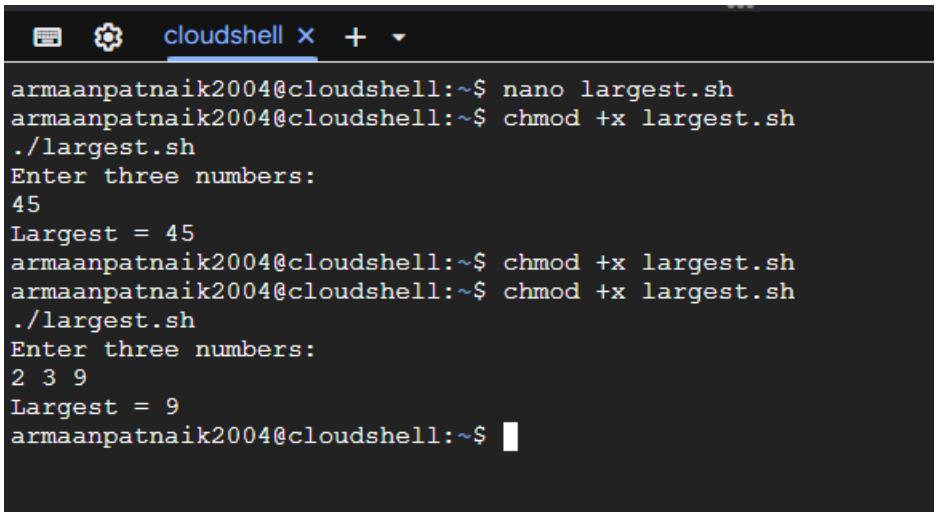
20. Even/Odd

Write a shell script that checks whether a number is even or odd.



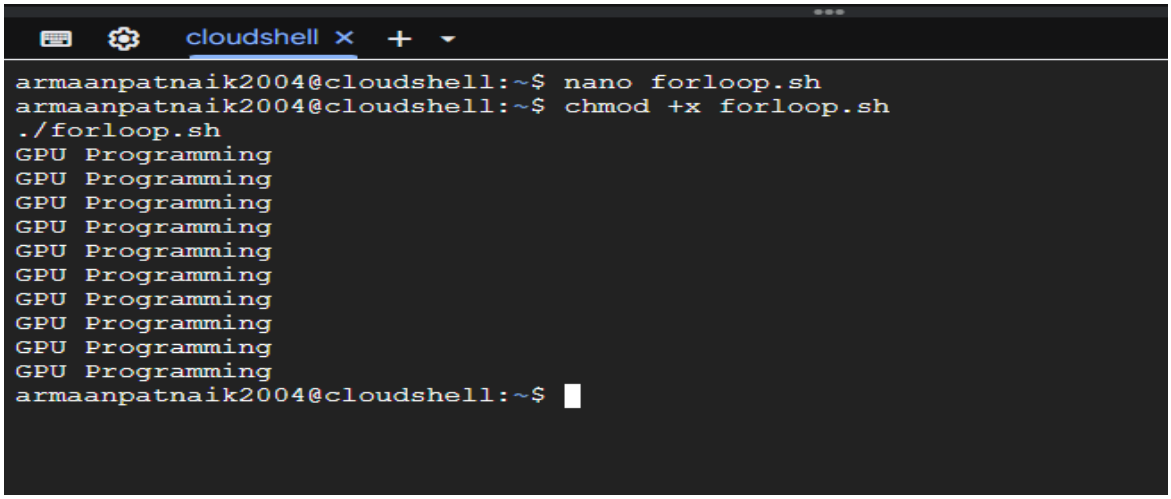
21. Largest Number

Write a shell script that finds the largest among three numbers.



Q.22. For Loop

Write a shell script that prints GPU Programming 10 times.



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Q.23. While Loop

Write a shell script that prints numbers from

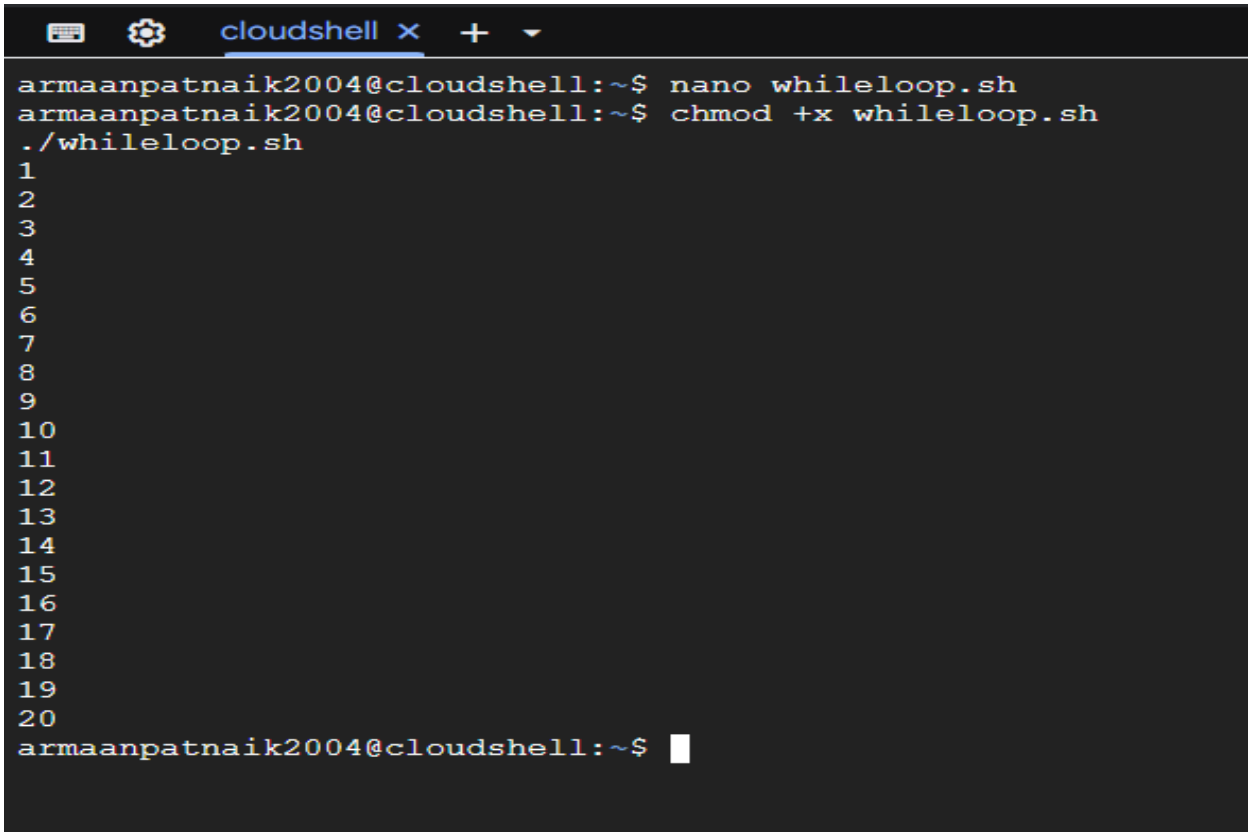
1 to 20

using a while loop. 23. While Loop

Write a shell script that prints numbers from

1 to 20

using a while loop.



Q. 24. File Existence

Write a script that checks whether a file exists.

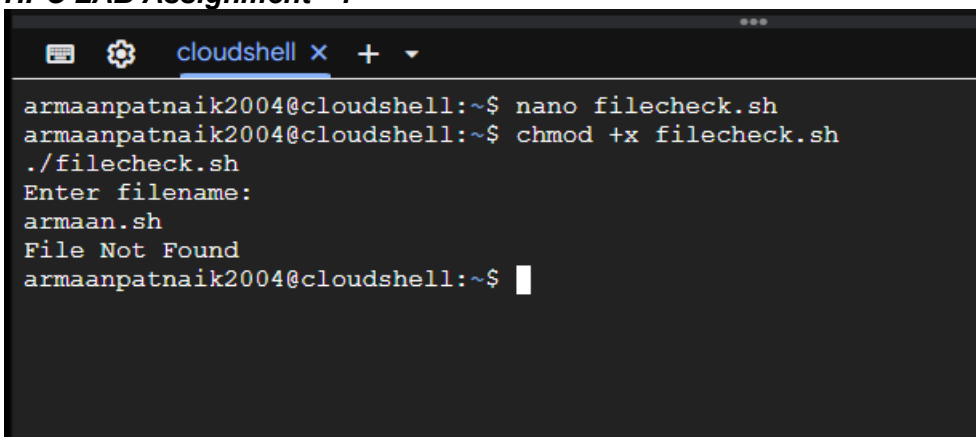
If yes, print

File Found

Otherwise print

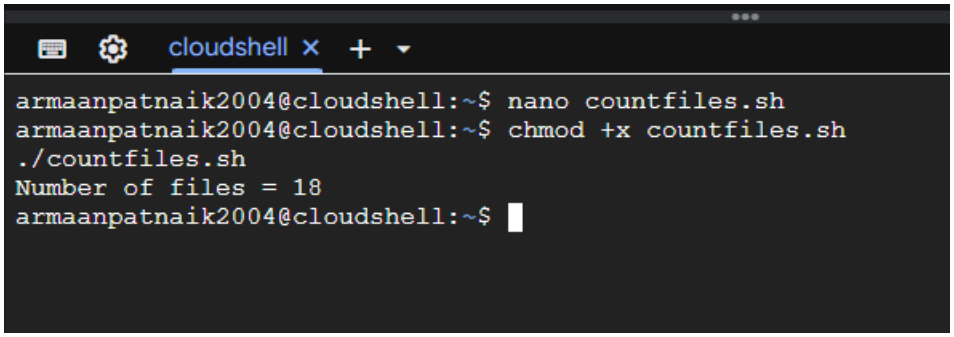
File Not Found

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25. Counting Files

Write a shell script that counts the number of files present in the current directory.



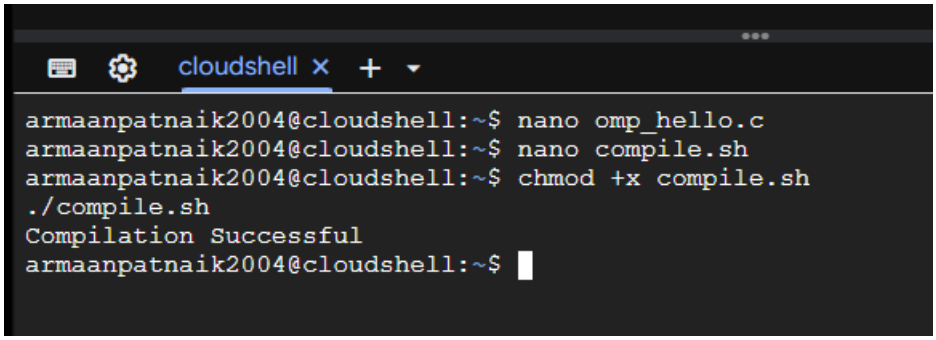
26. Compilation Script

Write a shell script that

compiles an OpenMP program using gcc

checks whether compilation was successful

displays an appropriate message



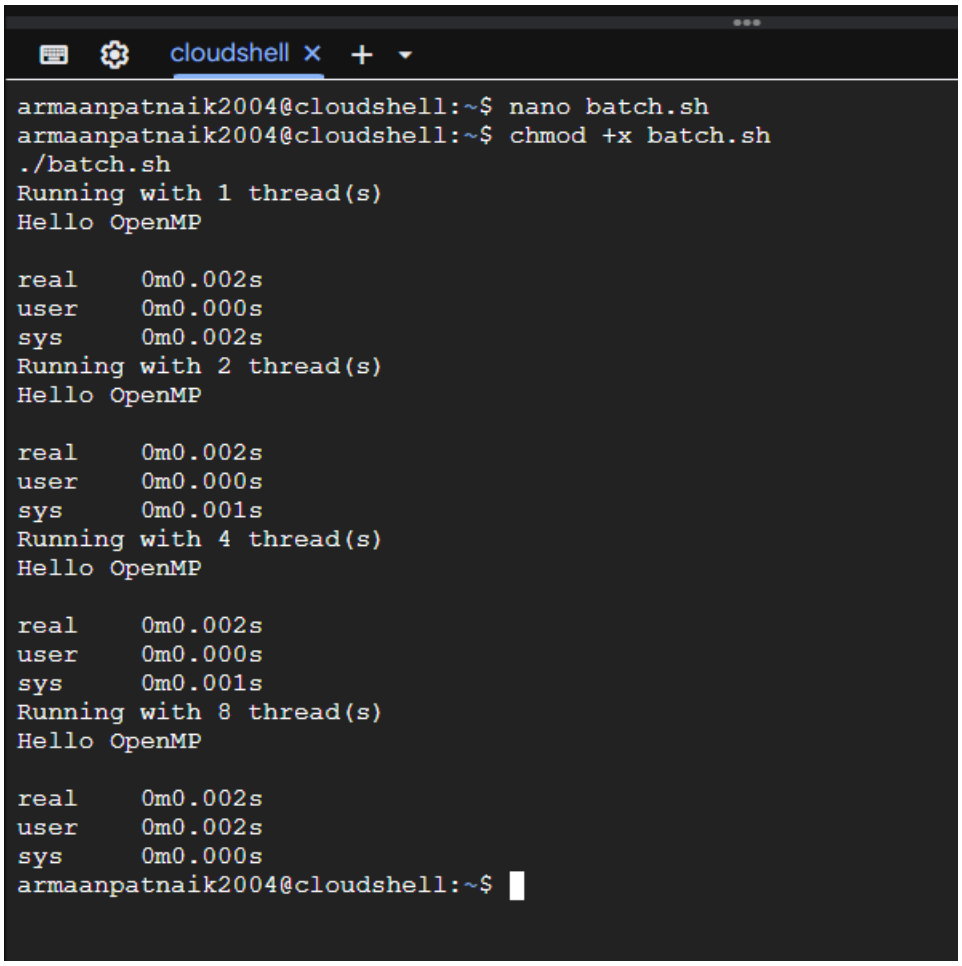
27. Batch Execution

Write a shell script that runs the same executable for thread counts

1

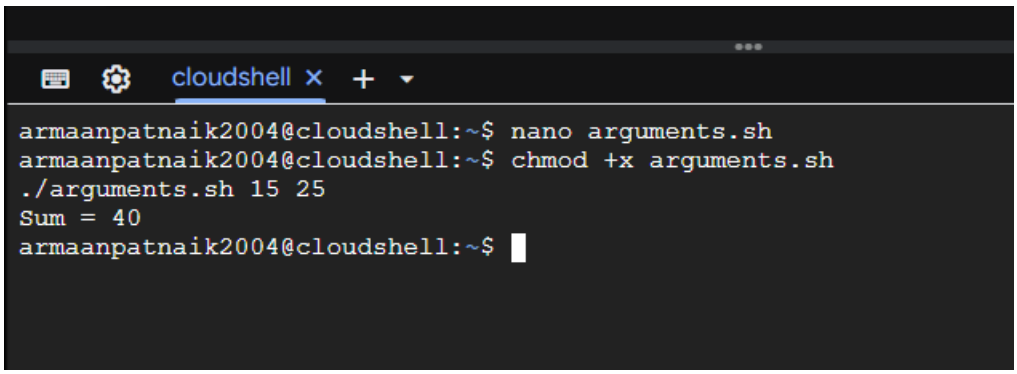
2

and prints the execution time for each run.



28. Command Line Arguments

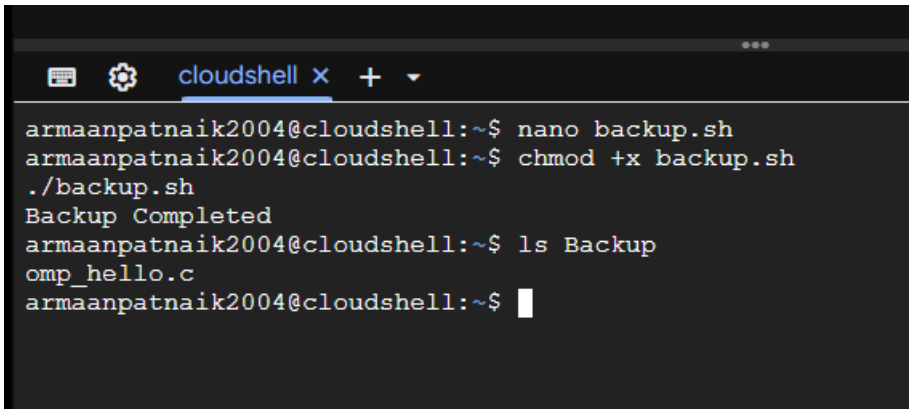
Write a shell script that accepts two numbers as command-line arguments and prints their sum.



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29. Backup Script

Write a shell script that creates a backup of all .c files into a folder named Backup.



30. Mini Project

Create a shell script called project_setup.sh that automatically:

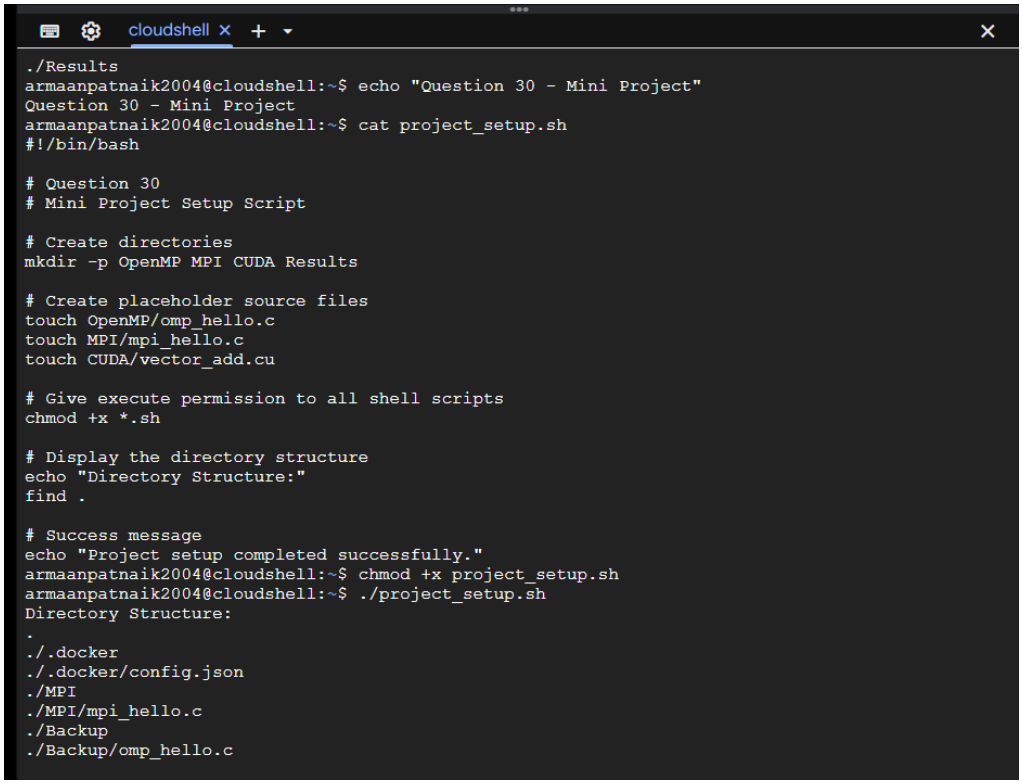
Creates the folders OpenMP, MPI, CUDA, and Results

Creates placeholder source files in each folder

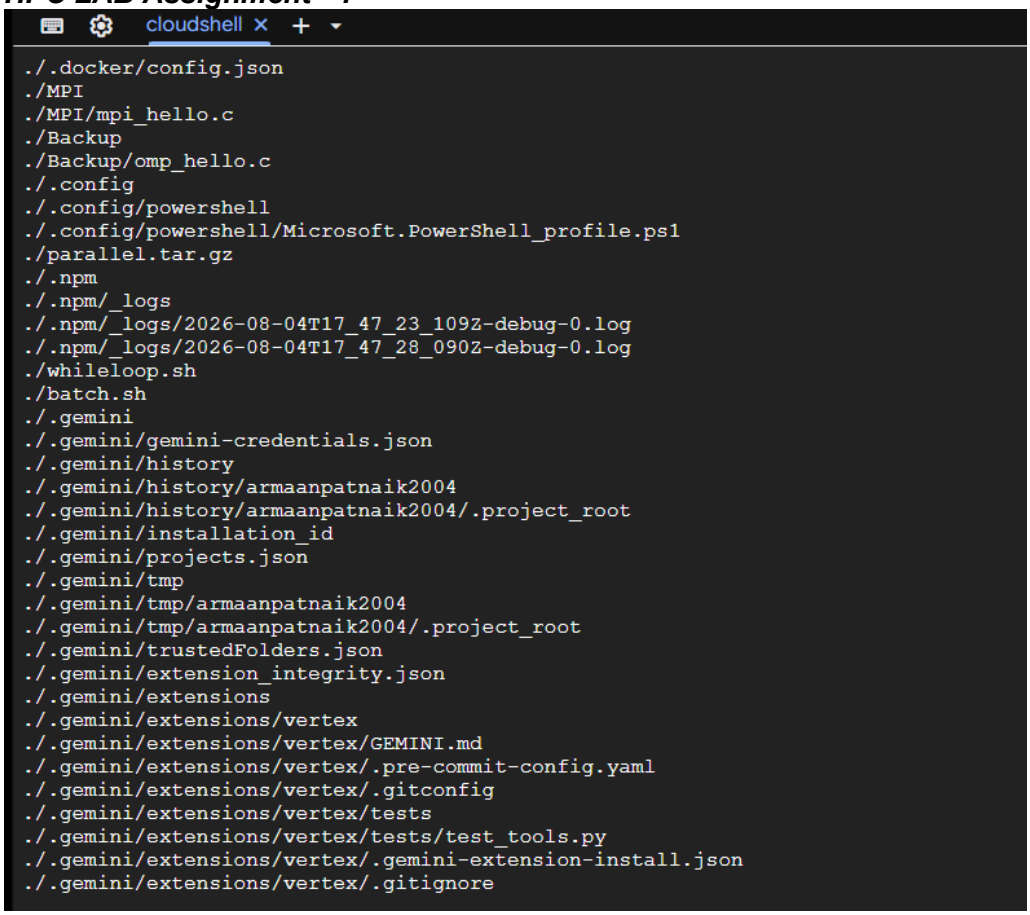
Gives execute permission to all .sh files

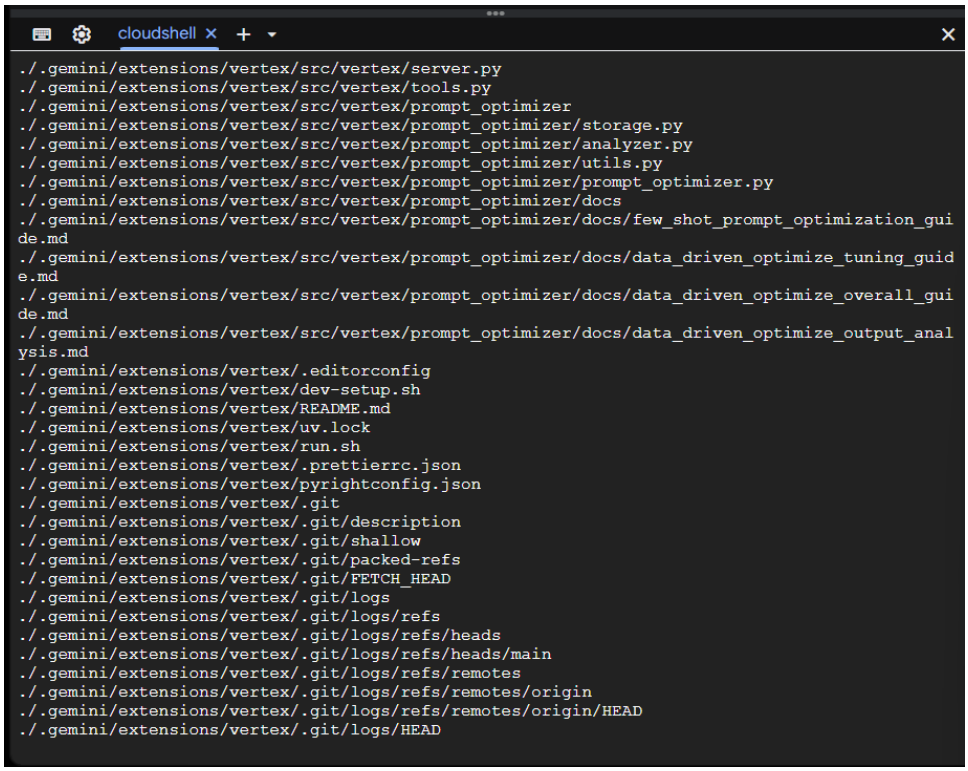
Displays the created directory structure

Prints "Project setup completed successfully."



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